

Chapter 1

Introduction to Probability

1.2 Interpretations of Probability

Commentary

It is interesting to have the students determine some of their own subjective probabilities. For example, let X denote the temperature at noon tomorrow outside the building in which the class is being held. Have each student determine a number x_1 such that the student considers the following two possible outcomes to be equally likely: $X \leq x_1$ and $X > x_1$. Also, have each student determine numbers x_2 and x_3 (with $x_2 < x_3$) such that the student considers the following three possible outcomes to be equally likely: $X \leq x_2$, $x_2 < X < x_3$, and $X \geq x_3$. Determinations of more than three outcomes that are considered to be equally likely can also be made. The different values of x_1 determined by different members of the class should be discussed, and also the possibility of getting the class to agree on a common value of x_1 .

Similar determinations of equally likely outcomes can be made by the students in the class for quantities such as the following ones which were found in the 1973 World Almanac and Book of Facts: the number of freight cars that were in use by American railways in 1960 (1,690,396), the number of banks in the United States which closed temporarily or permanently in 1931 on account of financial difficulties (2,294), and the total number of telephones which were in service in South America in 1971 (6,137,000).

1.4 Set Theory

Solutions to Exercises

1. Assume that $x \in B^c$. We need to show that $x \in A^c$. We shall show this indirectly. Assume, to the contrary, that $x \in A$. Then $x \in B$ because $A \subset B$. This contradicts $x \in B^c$. Hence $x \in A$ is false and $x \in A^c$.
2. First, show that $A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C)$. Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$. That is, $x \in A$ and either $x \in B$ or $x \in C$ (or both). So either $(x \in A \text{ and } x \in B)$ or $(x \in A \text{ and } x \in C)$ or both. That is, either $x \in A \cap B$ or $x \in A \cap C$. This is what it means to say that $x \in (A \cap B) \cup (A \cap C)$. Thus $A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C)$. Basically, running these steps backwards shows that $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$.
3. To prove the first result, let $x \in (A \cup B)^c$. This means that x is not in $A \cup B$. In other words, x is neither in A nor in B . Hence $x \in A^c$ and $x \in B^c$. So $x \in A^c \cap B^c$. This proves that $(A \cup B)^c \subset A^c \cap B^c$. Next, suppose that $x \in A^c \cap B^c$. Then $x \in A^c$ and $x \in B^c$. So x is neither in A nor in B , so it can't be in $A \cup B$. Hence $x \in (A \cup B)^c$. This shows that $A^c \cap B^c \subset (A \cup B)^c$. The second result follows from the first by applying the first result to A^c and B^c and then taking complements of both sides.

4. To see that $A \cap B$ and $A \cap B^c$ are disjoint, let $x \in A \cap B$. Then $x \in B$, hence $x \notin B^c$ and so $x \notin A \cap B^c$. So no element of $A \cap B$ is in $A \cap B^c$, hence the two events are disjoint. To prove that $A = (A \cap B) \cup (A \cap B^c)$, we shall show that each side is a subset of the other side. First, let $x \in A$. Either $x \in B$ or $x \in B^c$. If $x \in B$, then $x \in A \cap B$. If $x \in B^c$, then $x \in A \cap B^c$. Either way, $x \in (A \cap B) \cup (A \cap B^c)$. So every element of A is an element of $(A \cap B) \cup (A \cap B^c)$ and we conclude that $A \subset (A \cap B) \cup (A \cap B^c)$. Finally, let $x \in (A \cap B) \cup (A \cap B^c)$. Then either $x \in A \cap B$, in which case $x \in A$, or $x \in A \cap B^c$, in which case $x \in A$. Either way $x \in A$, so every element of $(A \cap B) \cup (A \cap B^c)$ is also an element of A and $(A \cap B) \cup (A \cap B^c) \subset A$.
5. To prove the first result, let $x \in (\cup_i A_i)^c$. This means that x is not in $\cup_i A_i$. In other words, for every $i \in I$, x is not in A_i . Hence for every $i \in I$, $x \in A_i^c$. So $x \in \cap_i A_i^c$. This proves that $(\cup_i A_i)^c \subset \cap_i A_i^c$. Next, suppose that $x \in \cap_i A_i^c$. Then $x \in A_i^c$ for every $i \in I$. So for every $i \in I$, x is not in A_i . So x can't be in $\cup_i A_i$. Hence $x \in (\cup_i A_i)^c$. This shows that $\cap_i A_i^c \subset (\cup_i A_i)^c$. The second result follows from the first by applying the first result to A_i^c for $i \in I$ and then taking complements of both sides.
6. (a) Blue card numbered 2 or 4.
 (b) Blue card numbered 5, 6, 7, 8, 9, or 10.
 (c) Any blue card or a red card numbered 1, 2, 3, 4, 6, 8, or 10.
 (d) Blue card numbered 2, 4, 6, 8, or 10, or red card numbered 2 or 4.
 (e) Red card numbered 5, 7, or 9.
7. (a) These are the points not in A , hence they must be either below 1 or above 5. That is $A^c = \{x : x < 1 \text{ or } x > 5\}$.
 (b) These are the points in either A or B or both. So they must be between 1 and 5 or between 3 and 7. That is, $A \cup B = \{x : 1 \leq x \leq 7\}$.
 (c) These are the points in B but not in C . That is $BC^c = \{x : 3 < x \leq 7\}$. (Note that $B \subset C^c$.)
 (d) These are the points in none of the three sets, namely $A^c B^c C^c = \{x : 0 < x < 1 \text{ or } x > 7\}$.
 (e) These are the points in the answer to part (b) and in C . There are no such values and $(A \cup B)C = \emptyset$.
8. Blood type A reacts only with anti-A, so type A blood corresponds to $A \cap B^c$. Type B blood reacts only with anti-B, so type B blood corresponds to $A^c B$. Type AB blood reacts with both, so $A \cap B$ characterizes type AB blood. Finally, type O reacts with neither antigen, so type O blood corresponds to the event $A^c B^c$.
9. (a) For each n , $B_n = B_{n+1} \cup A_n$, hence $B_n \supset B_{n+1}$ for all n . For each n , $C_{n+1} \cap A_n = C_n$, so $C_n \subset C_{n+1}$.
 (b) Suppose that $x \in \cap_{n=1}^{\infty} B_n$. Then $x \in B_n$ for all n . That is, $x \in \cup_{i=n}^{\infty} A_i$ for all n . For $n = 1$, there exists $i \geq 1$ such that $x \in A_i$. Assume to the contrary that there are at most finitely many i such that $x \in A_i$. Let m be the largest such i . For $n = m + 1$, we know that there is $i \geq n$ such that $x \in A_i$. This contradicts m being the largest i such that $x \in A_i$. Hence, x is in infinitely many A_i . For the other direction, assume that x is in infinitely many A_i . Then, for every n , there is a value of $j > n$ such that $x \in A_j$, hence $x \in \cup_{i=n}^{\infty} A_i = B_n$ for every n and $x \in \cap_{n=1}^{\infty} B_n$.
 (c) Suppose that $x \in \cup_{n=1}^{\infty} C_n$. That is, there exists n such that $x \in C_n = \cap_{i=n}^{\infty} A_i$, so $x \in A_i$ for all $i \geq n$. So, there are at most finitely many i (a subset of $1, \dots, n-1$) such that $x \notin A_i$. Finally, suppose that $x \in A_i$ for all but finitely many i . Let k be the last i such that $x \notin A_i$. Then $x \in A_i$ for all $i \geq k+1$, hence $x \in \cap_{i=k+1}^{\infty} A_i = C_{k+1}$. Hence $x \in \cup_{n=1}^{\infty} C_n$.

10. (a) All three dice show even numbers if and only if all three of A , B , and C occur. So, the event is $A \cap B \cap C$.
- (b) None of the three dice show even numbers if and only if all three of A^c , B^c , and C^c occur. So, the event is $A^c \cap B^c \cap C^c$.
- (c) At least one die shows an odd number if and only if at least one of A^c , B^c , and C^c occur. So, the event is $A^c \cup B^c \cup C^c$.
- (d) At most two dice show odd numbers if and only if at least one die shows an even number, so the event is $A \cup B \cup C$. This can also be expressed as the union of the three events of the form $A \cap B \cap C^c$ where exactly one die shows odd together with the three events of the form $A \cap B^c \cap C^c$ where exactly two dice show odd together with the even $A \cap B \cap C$ where no dice show odd.
- (e) We can enumerate all the sums that are no greater than 5: $1 + 1 + 1$, $2 + 1 + 1$, $1 + 2 + 1$, $1 + 1 + 2$, $2 + 2 + 1$, $2 + 1 + 2$, and $1 + 2 + 2$. The first of these corresponds to the event $A_1 \cap B_1 \cap C_1$, the second to $A_2 \cap B_1 \cap C_1$, etc. The union of the seven such events is what is requested, namely

$$(A_1 \cap B_1 \cap C_1) \cup (A_2 \cap B_1 \cap C_1) \cup (A_1 \cap B_2 \cap C_1) \cup (A_1 \cap B_1 \cap C_2) \cup (A_2 \cap B_2 \cap C_1) \cup (A_2 \cap B_1 \cap C_2) \cup (A_1 \cap B_2 \cap C_2).$$

11. (a) All of the events mentioned can be determined by knowing the voltages of the two subcells. Hence the following set can serve as a sample space

$$S = \{(x, y) : 0 \leq x \leq 5 \text{ and } 0 \leq y \leq 5\},$$

where the first coordinate is the voltage of the first subcell and the second coordinate is the voltage of the second subcell. Any more complicated set from which these two voltages can be determined could serve as the sample space, so long as each outcome could at least hypothetically be learned.

- (b) The power cell is functional if and only if the sum of the voltages is at least 6. Hence, $A = \{(x, y) \in S : x + y \geq 6\}$. It is clear that $B = \{(x, y) \in S : x = y\}$ and $C = \{(x, y) \in S : x > y\}$. The powercell is not functional if and only if the sum of the voltages is less than 6. It needs less than one volt to be functional if and only if the sum of the voltages is greater than 5. The intersection of these two is the event $D = \{(x, y) \in S : 5 < x + y < 6\}$. The restriction “ $\in S$ ” that appears in each of these descriptions guarantees that the set is a subset of S . One could leave off this restriction and add the two restrictions $0 \leq x \leq 5$ and $0 \leq y \leq 5$ to each set.
- (c) The description can be worded as “the power cell is not functional, and needs at least one more volt to be functional, and both subcells have the same voltage.” This is the intersection of A^c , D^c , and B . That is, $A^c \cap D^c \cap B$. The part of D^c in which $x + y \geq 6$ is not part of this set because of the intersection with A^c .
- (d) We need the intersection of A^c (not functional) with C^c (second subcell at least as big as first) and with B^c (subcells are not the same). In particular, $C^c \cap B^c$ is the event that the second subcell is strictly higher than the first. So, the event is $A^c \cap B^c \cap C^c$.

1.5 The Definition of Probability

Solutions to Exercises

1. Define the following events:

$$\begin{aligned} A &= \{\text{the selected ball is red}\}, \\ B &= \{\text{the selected ball is white}\}, \\ C &= \{\text{the selected ball is either blue, yellow, or green}\}. \end{aligned}$$